



One-shot adaptation for cross-domain semantic segmentation in remote sensing images

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ABSTRACT

Contemporary cross-domain remote sensing (RS) image segmentation has been successful in recent years. When the target domain data becomes scarce in some realistic scenarios, the performance of traditional domain adaptation (DA) methods significantly drops. In this paper, we tackle the problem of fast cross-domain adaptation by observing only one unlabeled target data. To deal with dynamic domain shift efficiently, this paper introduces a novel framework named Minimax One-shot Adaptation (MOAT) to perform cross-domain feature alignment in semantic segmentation. Specifically, MOAT alternately maximizes the cross-entropy to select the most informative source samples and minimizes the cross-entropy of obtained samples to make the model fit the target data. The selected source samples can effectively describe the target data distribution using the proposed uncertainty-based distribution estimation technique. We propose a memory-based feature enhancement strategy to learn domain-invariant decision boundaries to accomplish semantic alignment. Generally, we empirically demonstrate the effectiveness of the proposed MOAT. It achieves a new state-of-the-art performance on cross-domain RS image segmentation for conventional unsupervised domain adaptation and one-shot domain adaptation scenarios.

1. Introduction

Remote sensing (RS) technology, referring to the acquisition of information about the Earth's surface using sensors, has been broadly applied in various real-world tasks, including remote sensing scene classification, object detection, and semantic segmentation. Among these applications, semantic segmentation has achieved remarkable progress with vast amounts of pixel-level annotated training data. However, collecting these large-scale datasets requires tremendous labeling efforts in both time and cost. To reduce this burden of labeling, previous cross-domain semantic segmentation [1–3] has been developed by utilizing and transferring the knowledge of label-rich realistic synthetic data (source) to the unlabeled real-world data (target), which is referred as domain adaptive semantic segmentation (DASS). The essential goal of DASS is to eliminate domain shift, and existing DASS methods can be roughly divided into two categories, *i.e.*, the adversarial-learning-based methods [1,2,4] and the self-training-based methods [3,5,6].

The former focuses on aligning the distributions of source and target domains at image-level, feature-level, and output-level. The latter aims to retrain the model utilizing the generated pseudo labels in the target domain.

Despite significant efforts in developing DASS methods, most of them require large amounts of target data and hundreds of training epochs, which is inflexible and inefficient for handling domain adaptation with dynamic domain shift problems. Although some approaches attempt to few-shot scenarios [7–10], there are still some issues existing. For example, it is difficult to obtain sufficient target images in RS due to the high requirement of sensors and data privacy concerns. To address this problem, we propose and explore an efficient one-shot adaptation problem that enables quick adaptation from the source domain to a new target domain by only one target data accessible and a few training iterations. It is practical and efficient for real-world applications, but the scarcity of target data and the requirement of

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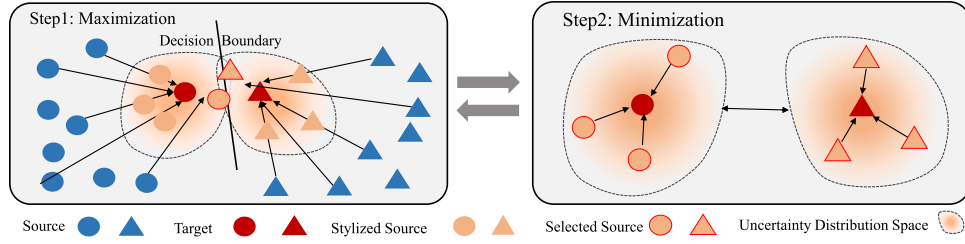


Fig. 1. Our model learns discriminative features for target data by the following two alternate steps: (1) we select the most informative samples with the target style, and (2) we minimize the cross-entropy for the selected images to learn decision boundaries fitting to the target data. The circle and triangle denote different categories.

quickly generalizing to different scenarios also bring more challenges.

Compared to conventional domain adaptation, one-shot domain adaptation is realistic and practical from the following aspects. (1) Although any camera mounted on an automobile can capture unlabeled target images, in some privacy scenarios, an extensive collection of unlabeled target data may be unavailable or difficult to obtain, especially for remote sensing images. (2) RS images from different domains have large visual differences and images contain complex geographical information, resulting in the direct adversarial learning process suffering from instability. Therefore, it is critical for a model to address the dynamic domain shift quickly and obtain better performance in the real world. (3) One-shot domain adaptation [11–13] provides a research direction of efficiency, which satisfies the requirements of quick adaptation for addressing dynamic domain shift problems. To address semantic segmentation for Remote Sensing (RS) images, previous works [14,15] have explored data-efficient methods in few-shot and one-shot scenarios. Specifically, CPM [14] introduces a multiple perturbation dynamic consistency approach with class prototype memory for semi-supervised RS semantic segmentation, [16,17] focus on the cross-domain few-shot semantic segmentation on RS images using the prompt learning, while LRM_I2I [15] presents the first attempt at one-shot domain adaptation for RS images. Although this method achieves promising results for OSDA, the additional image-to-image translation module is not convenient due to the diversity and complexity of RS images.

To address above problems, we develop a Minimax One-shot Adaptation (MOAT) approach in this paper, which aims to achieve adaptation with only a few iterations. As illustrated in Fig. 1, our key idea is to alternately (1) maximize the cross-entropy to select the most informative source images that can describe the target distribution and (2) minimize the cross-entropy of obtained samples to make the model fit the target data. Specifically, in the maximization step, we first introduce a simple and effective uncertainty-based target distribution estimation technique, which can explicitly synthesize stylized source domain images in both image and feature levels according to the estimated diversity target style. Then, we perform an inference process with no gradient to select the most efficient images based on the stylized samples. Besides, in the minimization step, we fine-tune the segmentation network using the selected instances to make the model fit the target data. Furthermore, to accomplish semantic alignment, we propose a memory-based feature enhancement strategy to maintain category prototypes and to learn domain-invariant decision boundaries.

The contributions of our work can be summarized as follows:

1. We propose an efficient one-shot adaptation framework for cross-domain RS image segmentation, which aims to quickly adapt the trained source model to the target data with only one given target image.
2. We propose a novel minimax training method for one-shot domain adaptation, which treats this problem as an efficient sample selection learning.
3. We show the effectiveness of our method by achieving a new state-of-the-art performance on one-shot domain adaptive segmentation in remote sensing benchmarks.

The rest of this paper is organized as follows. Section 2 gives a brief review of related works about unsupervised domain adaptive semantic segmentation, cross-domain semantic segmentation in RS images and one-shot cross-domain adaptation for semantic segmentation. Section 3 presents the procedure of the proposed MOAT to handle the one-shot cross-domain adaptive semantic segmentation task on RS images. Section 4 shows the datasets, results, analysis and ablation studies of UDA and OSDA experiments in RS images based on LoveDA, Potsdam and Vaihingen datasets, and synthetic-to-real scenarios based on GTA5 SYNTHIA to Cityscapes. Section 5 concludes the paper.

2. Related works

Unsupervised Domain Adaptive Semantic Segmentation aims to transfer the pixel-level annotations from the source domain to the target domain [18,19]. Existing approaches can be roughly categorized into two groups: adversarial learning based method and self-training based method. For adversarial learning, numerous works focus on reducing the distribution misalignment in the image level [2,20], feature level [21–23], and output level [1,4,24]. For the self-training based method, the essential idea is to generate reliable pseudo labels. Typical approaches usually consist of two steps: (1) generate pseudo labels based on the source model [3,5,25] or the learned domain-invariant model [6,26], (2) refine the target model supervised by the generated pseudo labels. PLCA [27] and S2T [28] introduce a new paradigm for domain adaptive semantic segmentation via building pixel-level cycle association between source and target pixel pairs. Moreover, recent methods [29–32] learn a super pre-trained network to transfer knowledge into the downstream tasks.

Cross-domain Semantic Segmentation on RS Images. Different from domain adaptive semantic segmentation on the natural scene images, RS images contain more specific detailed information like element corners, irregular shapes, etc. Specifically, Benjdira et al. [33] first introduce domain adaptation for the RS image segmentation and adopt GAN based architecture to align source and target images. Following this pioneering work, GAN-based networks have been explored to address domain adaptation for RS image semantic segmentation [34,35]. Moreover, inspired by the development of adversarial learning for domain adaptive semantic segmentation, existing methods [36,37] utilize adversarial learning to extract domain-invariant features. Another attempt is a curriculum-style approach [38,39], which proposes a local-to-global RS segmentation framework including “easy-to-adapt” and “hard-to-adapt” stages.

One-shot Cross-Domain Adaptation for Semantic Segmentation is a challenging and practical task in which only one single unlabeled target image is accessible. To address this data-scarce issue, one-shot domain adaptation (OSDA) methods [11–13,40] have been provided to achieve adaptation in an effective way. Specifically, ASM [11] estimates the target distribution by adopting the style transfer techniques. DATUM [13] tackles the target data-scarce problem by leveraging text-to-image models to generate photo-realistic synthetic target images. IDM [12] provides an efficient technique to handle the one-shot domain

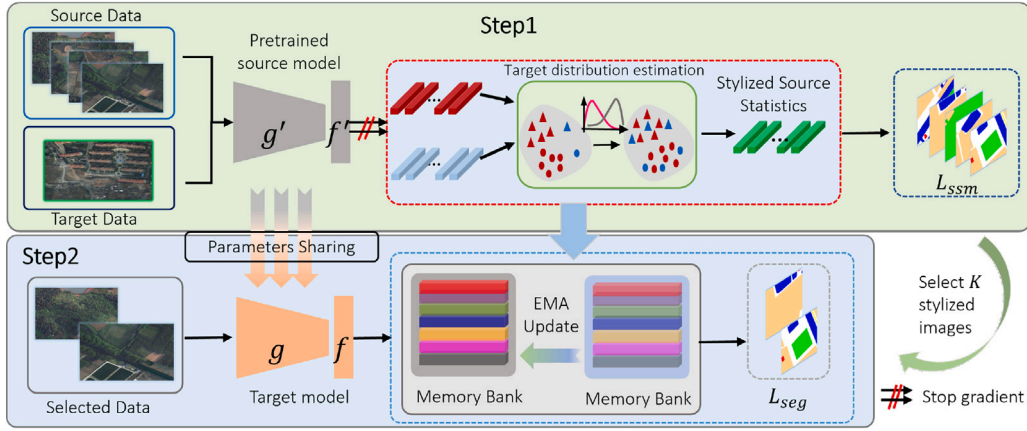


Fig. 2. An overview of our proposed MOAT. Step 1: We first use one target image and a subset of source images to generate target-style images via the target distribution estimation module. Then, we select K stylized images with the maximal cross-entropy loss. Step 2: Selected images are used to fine-tune the target segmentation model. We also propose a memory-augmented prototype learning module to improve the segmentation performance. g and f indicate the feature extractor and classifier, respectively.

adaptation problem by introducing an informative data mining method, which provides an insightful idea for exploring the information hidden in the source data. LRM_I2I [15] is the first work for one-shot domain adaptation on RS images. Different from LRM_I2I requiring an extra image-to-image translation paradigm, our method is more effective by only selecting the informative images on the stylized source data.

3. Method

In this section, we formally introduce the proposed Minimax One-shot AdapTation (**MOAT**) for RS semantic segmentation. In our scenario, we only access one unlabeled target image $x_t \in \mathcal{X}_t$, and n_s source data $\{x_s^i, y_s^i\}_{i=1}^{n_s} = \{\mathcal{X}_s, \mathcal{Y}_s\}$. The overall pipeline is shown in Fig. 2.

3.1. Minimax adaptation

Given a well-trained source model $\mathcal{M}_s$, our goal is to obtain a target model $\mathcal{M}$ by adapting it to the target domain quickly. Due to the lack of statistical support for the style and content of single-target domain samples, the problem evolves into iteratively selecting the most informative samples from the labeled source domain. Intuitively, a high-quality sample should have two properties: (1) it should fit the overall statistical distribution of the target domain as closely as possible, and (2) it is effective for learning the target decision boundary.

Guided by the above, we treat the solely given target style as an “anchor style” similar to [11]. Then we introduce a target distribution estimation strategy to generate diverse stylized source images that are similar to the target data. We select the most informative stylized images by performing cross-entropy maximization with respect to the efficient samples. Then, we extract the discriminative features by performing cross-entropy minimization with respect to optimize the segmentation model. We hypothesize that selected representative samples with the target-like style can describe the target distribution and minimize the cross-entropy of these samples, which is conducive to fast convergence.

Therefore, we hope to learn a sample-efficient model that can quickly adapt the source model to the target domain and give an alternate minimax optimization paradigm as below:

$$\text{Step 1: } \mathcal{L}_{ssm} = \arg\max_{k \in n_s} \sum_{k=1}^K W_k, \quad (1)$$

$$\text{Step 2: } \mathcal{L}_{seg} = -\frac{1}{K} \sum_{k=1}^K \sum_{j=1}^{H \times W} y_s^{k,j} \log(p(\hat{x}_s^{k,j})), \quad (2)$$

where **Step 1** is an inference process that selects K efficient samples from n_s stylized source images according to the cross-entropy metric W .

Then, **Step 2** utilizes the selected K images to fine-tune the segmentation model. $p(\hat{x}_s^{k,j})$ indicates the probabilistic output of the j th pixel for the k th image $\hat{x}_s$ and $p(x) = g \circ f(x)$, where g and f indicate the feature extractor and classifier. $(\hat{x}_s, y_s)$ denotes a stylized image training pair with corresponding ground truth label y_s . (H, W) is the size of training images.

Target Distribution Estimation To estimate target distribution, we introduce an uncertainty-based domain shift method to diversify the target style and alleviate the problem of poor distribution of a single target sample. We treat the style of a given target image as “anchor style” that provides a certain direction and intensity for estimating the distribution of the target domain. A query image is randomly selected from the source domain. The domain shift is represented by the feature statistic offset of source and target data. To diversify the target style, we randomly sample the offset scale of each pair from a Beta distribution, which corresponds to uncertainty domain shift, inspired by [41,42]. The reconstructed feature statistic of mean $\gamma(x_t)$ and standard deviation $\beta(x_t)$ are:

$$\begin{aligned} \gamma(x_t) &= \mu(x_t) + \delta_\mu \|(\mu(x_t) - \mu(x_s))\|, \\ \beta(x_t) &= \sigma(x_t) + \delta_\sigma \|(\sigma(x_t) - \sigma(x_s))\|, \end{aligned} \quad (3)$$

where δ_μ and δ_σ control the weights of uncertainty factor and $\delta_\mu, \delta_\sigma \in \mathcal{B}(\lambda_B, \lambda_B)$ are randomly sampled from Beta distribution with the hyper-parameter λ_B . Once we obtained the target distribution with uncertainty, the stylized feature of source images can be calculated as follows:

$$\hat{x}_s = \beta(x_t) \left(\frac{x_s - \mu(x_s)}{\sigma(x_s)} \right) + \gamma(x_t). \quad (4)$$

Note that the proposed strategy can be integrated at various positions of the networks, including the original image level and each layer of feature extractor. We can successfully address the style overfitting problem by generating various target-style synthetic images that represent the distribution of the target domain, which are referred to as stylized images in this paper.

Sample Selection in Maximization After generating stylized images, the fast adaptation evolves into an effective sample selection problem in choosing the most informative images that enable quick transformation from the source domain to dynamic target domains. To achieve this, we explore samples close to the source decision boundaries by leveraging the cross-entropy maximization. These stylized images with maximal cross-entropy values are usually the most informative samples to optimize the target model. Consequently, the sample selection strategy in the maximization is hereby introduced to pick out K samples as the

following formulation:

$$\mathcal{L}_{ssm} = \operatorname{argmax} \sum_{k=1}^K (\mathbb{E}_{(\hat{x}_s, y_s) \in (\mathcal{X}_s, \mathcal{Y}_s)} y_s \log(\hat{p}(\hat{x}_s))), \quad (5)$$

where K is the number of selected samples with the maximal cross-entropy values from n_s source stylized images. This is an inference process that does not need backpropagation. $\hat{p}(x)$ denotes the probability output with no gradient. Instead of inferring all source images, we randomly choose a subset with $n_s = 100$ samples in our experiments to save search time.

3.2. Memory enhanced prototype in minimization

Domain-invariant structure information is another essential prior knowledge to fast model adaptation. Despite the domain discrepancy, semantically identical structure information is consistent between source and target data, e.g., the representation of class prototypes. We hope to use the limited context of the target domain single-shot image to establish an adequate semantic association with the source domain data. Therefore we construct a memory bank to store the semantic prototype of the source domain data to enhance discriminative feature learning. According to the existing works [6,43], we first calculate the initialized prototype for i th category on the source images,

$$c_i = \frac{\sum_{x_s \in \mathcal{X}_s} f' \circ g'(x_s)^i \times \mathbb{I}(y_s^i = 1)}{\sum_{x_s \in \mathcal{X}_s} \mathbb{I}(y_s^i = 1)}, \quad (6)$$

where $\mathbb{I}$ is the indicator function, and f' and g' are the feature extractor and classifier from the trained source model. The dimension for the prototype c_i is $\mathbf{R}^{1 \times 256}$. Note that this initialized prototype is attended based on the trained source model without gradient propagation. Then the memory is updated by the feature of target-stylized images as $c = mc + (1 - m)c_t$, where m is a momentum coefficient and c_t is the prototype computed from the target stylized image.

Domain-invariant prototype provides a structure discriminative information, which pulls features with the same semantics closer and pushes different semantic features further. The final cross-entropy minimization in Eq. (2) is reformulated as revealing the confidence of classification,

$$\mathcal{L}_{seg} = -\frac{1}{K} \sum_{k=1}^K \sum_{j=1}^{H \times W} y_s^{k,j} \log(w(\hat{x}_s^{k,j}) * p(\hat{x}_s^{k,j})), \quad (7)$$

where $p(\hat{x}_s^{k,j}) \in \mathbf{R}^{1 \times 19}$ represents the softmax prediction of $\hat{x}_s^{k,j}$ and w represents classification weights obtained from the memory-enhanced prototype:

$$w(\hat{x}_s^{k,j}) = \frac{\exp(-\|g(\hat{x}_s^{k,j}) - c\|/\tau)}{\sum_{c=1}^C \exp(-\|g(\hat{x}_s^{k,j}) - c\|/\tau)}, \quad (8)$$

where C is the number of classes, τ is the temperature, and $g(\hat{x}_s^{k,j}) \in \mathbf{R}^{N \times 256 \times H \times W}$ is the feature map from classifier.

3.3. Algorithm

The algorithm of the proposed Minimax One-shot AdapTation (MOAT) is illustrated in Algorithm 1.

4. Model results and analysis

4.1. Datasets and metric

We evaluate our method on two different domain adaptation semantic segmentation scenarios. The first is RS image semantic segmentation based on benchmark datasets: LoveDA [44], Potsdam [45], and Vaihingen [45]. Another is synthetic-to-real cross-domain semantic segmentation based on GTA5 [46] and Cityscapes [47].

Algorithm 1: The algorithm of the proposed method MOAT.

Data: Source images $(x_s, y_s) \in \{\mathcal{X}_s, \mathcal{Y}_s\}$, one unlabeled target image x_t , hyper-parameters λ_B , K , and m .

Result: The output target semantic segmentation model $\mathcal{M}$.

- 1 **Initialization:**
- 2 $\mathcal{M} \leftarrow \mathcal{M}_s$;
- 3 Calculate the category-level prototype c :

$$c_i = \frac{\sum_{x_s \in \mathcal{X}_s} f' \circ g'(x_s)^i \times \mathbb{I}(y_s^i = 1)}{\sum_{x_s \in \mathcal{X}_s} \mathbb{I}(y_s^i = 1)}.$$
- 4 **while** $iter \leq max_iter$ **do**
- 5 **Maximization Step:**
- 6 Compute the target feature statistic μ_t, σ_t as the anchor style;
- 7 **for** $m \leftarrow 0$ **to** $M = 100$ **do**
- 8 **Target Distribution Estimation:**
- 9 $\hat{x}_s = \beta(x_t)(\frac{x_s - \mu(x_s)}{\sigma(x_s)}) + \gamma(x_t)$.
- 10 **Sample Selection in Maximization:**
- 11 $\mathcal{L}_{seg} = -\frac{1}{K} \sum_{k=1}^K \sum_{j=1}^{H \times W} y_s^{k,j} \log(p(\hat{x}_s^{k,j}))$,
- 12 $\mathcal{L}_{ssm} = \operatorname{argmax} \sum_{k=1}^K (\mathbb{E}_{(\hat{x}_s, y_s) \in (\mathcal{X}_s, \mathcal{Y}_s)} y_s \log(\hat{p}(\hat{x}_s)))$.
- 13 **end**
- 14 **Minimization Step:**
- 15 **for** $k \leftarrow 0$ **to** K **do**
- 16 $\mathcal{L}_{seg} = -\frac{1}{K} \sum_{k=1}^K \sum_{j=1}^{H \times W} y_s^{k,j} \log(w(\hat{x}_s^{k,j}) * p(\hat{x}_s^{k,j}))$,
- 17 Update the prototype memory using the stylized feature.
- 18 **end**
- 19 **end**

LoveDA [44] is a recently proposed dataset for both RS semantic segmentation and domain adaptation tasks. It consists of 5987 RS images collected from three cities, including Nanjing, Changzhou, and Wuhan. It contains two different domains, including Urban and Rural. The Urban domain is split into 1156 training images and 677 validation images. For Rural domain, there are 1366 images for training and 992 images for validation. In this scenario, we design two cross-domain RS semantic segmentation tasks, which are Urban-to-Rural and Rural-to-Urban, and for each adaptation task, we have conducted experiments on both traditional domain adaptation and one-shot domain adaptation.

Potsdam and Vaihingen. These two datasets come from ISPRS 2D semantic segmentation benchmark dataset [45]. The Potsdam contains 38 VHR TOP (very-high-resolution True Orthophotos) images with a fixed size of 6000×6000 . There are three different modes in Potsdam, namely IR-R-G, R-G-B, and R-G-B-IR. The first two modes are 3-channel while the last mode is 4-channel. Vaihingen only has one imaging mode, which is IR-R-G. Therefore, in our experiments, we conduct adaptation on the IR-R-G mode both from Potsdam to Vaihingen and from Vaihingen to Potsdam. Following previous works [36,48–51], we crop the images of these two datasets into smaller sizes of 512×512 , with the cropping strides of 512×512 . Moreover, we split the Potsdam and Vaihingen datasets into training and testing sets. Specifically, there are 2904 and 1694 images for Potsdam in the training and testing set, respectively. For Vaihingen, the training and testing sets contain 1296 and 440 images.

Cityscapes, treated as target data, is a real-world dataset collected from several German cities. It has 2975 training images with a resolution of 2048×1024 . In our experiments, we use only one unlabeled image during training. We use the full validation set with 500 images to test our network.

GTA5, consisting of 24,966 images, is collected from the homonymous computer game, and the original image size is 1914×1052 . It has 19 common categories with Cityscapes, and the ground truth is generated by the game render itself.

Table 1

Comparison results for conventional cross-domain adaptation for RS image segmentation from Potsdam to Vaihingen. # TS denotes the number of target samples used in training. We use DeeplabV3 (ResNet-101) and SegFormer (Mit-B5) as our backbone. The best results are presented in **bold**.

Methods	#TS	Clutter		Imp. Surf.		Car		Tree		Low Vege.		Building		Overall	
		IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	mIoU	Ave. F1
AdaptSeg [1]	All	4.60	8.76	54.39	70.39	6.40	11.99	52.65	68.96	28.98	44.91	63.14	77.40	35.02	47.05
UncerDA [3]	All	8.74	–	58.30	–	32.21	–	53.43	–	39.57	–	55.39	–	41.27	–
FSDAN [48]	All	10.00	–	57.40	–	37.00	–	58.40	–	41.70	–	57.80	–	43.70	–
ProDA [6]	All	3.99	8.21	62.51	76.85	39.20	56.52	56.26	72.09	34.49	51.65	71.61	82.95	44.68	58.05
DualGAN [49]	All	29.66	45.65	49.41	66.13	34.34	51.09	57.66	73.14	38.87	55.97	62.30	76.77	45.38	61.43
Bai et al. [36]	All	19.60	32.80	65.00	78.80	39.60	56.70	54.80	70.80	36.20	53.20	76.00	86.40	48.50	63.10
Zhang et al. [51]	All	20.71	31.34	67.74	80.13	44.90	61.94	55.03	71.90	47.02	64.16	76.75	86.65	52.03	66.02
Wang et al. [55]	All	21.85	35.87	76.58	86.73	35.44	52.33	55.22	71.15	49.97	66.64	82.74	90.56	53.63	67.21
CIA-UDA [50]	All	27.80	43.51	63.28	77.51	52.91	69.21	64.11	78.13	48.03	64.90	75.13	85.80	55.21	69.84
ResiDualGAN [49]	All	11.64	18.42	72.29	83.89	57.01	72.51	63.81	77.88	49.69	66.29	80.57	89.23	55.83	68.04
MemoryAdapt [56]	All	50.30	–	79.65	–	61.09	–	78.52	–	68.22	–	86.45	–	56.05	–
MOAT (DeeplabV3)	All	53.88	84.71	80.07	90.50	63.61	82.41	51.11	53.22	51.29	84.60	89.39	93.18	64.89	81.44
SegFormer [57]	All	4.22	9.47	61.03	76.07	31.13	47.89	66.31	78.87	44.47	60.38	75.50	87.95	46.11	60.11
DAFormer [53]	All	48.26	60.17	74.09	84.12	38.96	56.41	70.88	81.36	57.53	71.48	84.07	90.75	62.30	74.05
ST-DASegNet [58]	All	67.03	80.28	74.43	85.36	43.38	60.49	67.36	80.49	48.57	65.37	85.23	92.03	64.33	77.34
PCEL [59]	All	63.69	–	75.80	–	71.32	–	78.42	–	58.28	–	93.12	–	73.44	–
MOAT (SegFormer)	All	69.22	89.52	82.11	88.64	50.05	95.07	77.74	95.33	59.99	64.78	90.55	97.13	71.61	88.41

Table 2

Comparison results for conventional cross-domain adaptation for RS image segmentation from Vaihingen to Potsdam. # TS denotes the number of target samples used in training. We use DeeplabV3 (ResNet-101) and SegFormer (Mit-B5) as our backbone. The best results are presented in **bold**.

Methods	# TS	Clutter		Imp. Surf.		Car		Tree		Low Vege.		Building		Overall	
		IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	mIoU	Ave. F1
AdaptSeg [1]	All	8.36	15.33	49.55	64.64	40.95	58.11	22.59	36.79	34.43	61.50	48.01	63.41	33.98	49.96
ProDA [6]	All	10.63	19.21	44.70	61.72	46.78	63.74	31.59	48.02	40.55	57.71	56.85	72.49	38.51	53.82
DualGAN [49]	All	11.48	20.56	51.01	67.53	48.49	65.31	34.98	51.82	36.50	53.48	53.37	69.59	39.30	54.71
Zhang et al. [51]	All	12.31	24.59	64.39	78.59	59.35	75.08	37.55	54.60	47.17	63.27	66.44	79.84	47.87	62.66
Wang et al. [55]	All	11.65	19.47	73.43	84.55	63.86	77.85	32.68	47.36	47.69	63.45	76.32	87.43	50.94	63.31
CIA-UDA [50]	All	10.87	19.61	62.74	77.11	65.35	79.04	47.74	64.63	54.40	70.47	72.31	83.93	52.23	65.80
MemoryAdapt [56]	All	16.15	–	79.70	–	76.54	–	64.06	–	67.58	–	77.84	–	49.82	–
MOAT (DeeplabV3)	All	13.61	24.00	61.71	68.94	76.45	90.62	48.01	55.13	56.04	70.42	71.63	91.20	54.58	66.68
SegFormer [57]	All	1.08	2.65	60.63	76.47	58.99	73.14	30.07	46.24	51.91	68.82	74.85	87.18	46.29	59.08
DAFormer [53]	All	2.56	5.02	68.42	79.07	65.20	79.31	70.65	82.13	56.39	72.48	78.94	87.64	57.03	67.61
ST-DASegNet [58]	All	0.18	0.35	76.45	86.65	73.54	84.76	62.89	77.22	61.04	75.80	83.81	91.19	59.65	69.33
PCEL [59]	All	21.53	–	65.04	–	76.95	–	70.96	–	63.09	–	82.64	–	63.37	–
MOAT (SegFormer)	All	6.52	6.75	81.64	91.11	76.05	93.46	66.13	75.87	68.19	86.01	88.42	96.92	64.49	75.02

4.2. Implementation details

We conduct all experiments by using **PyTorch** trained on the **NVIDIA A30 GPU**. In our experiments, we use different architectures and backbones to demonstrate the effectiveness of our method, including DeepLab-V2/V3 (ResNet-50/ResNet-101) and SegFormer (Mit-b5), which is the same as previous works [1,3,12,44,52,53]. We first pre-train the segmentation network on the source domain for 20k iterations to obtain a high-quality source model following [1]. Then it is considered as the initialization model. During one-shot adaptation, we follow previous works [11,12,54] using Stochastic Gradient Descent (SGD) optimizer with the learning rate 2.5×10^{-5} , momentum 0.9, and weight decay 5×10^{-4} . We schedule the learning rate using the “poly” policy: the learning rate is multiplied by $(1 - \frac{iter}{max_iter})^{0.9}$. The batch size is set to 1. Concerning the hyper-parameters, we utilize $m = 0.8$, $K = 3$, $\lambda_B = 0.5$, and $\tau = 1$ for all experiments. As for unsupervised domain adaptation, the implementation details are similar to the one-shot domain adaptation. Specifically, we use a Stochastic Gradient Descent (SGD) optimizer with the learning rate 2.5×10^{-5} , momentum 0.9, and weight decay 5×10^{-4} for training 20k iterations. We schedule the learning rate using the “poly” policy: the learning rate is multiplied by $(1 - \frac{iter}{max_iter})^{0.9}$. The batch size is set to 1. Different from one-shot scenarios, we use all unlabeled target images during training. Following [11], we run our methods 5 times with different random seeds to get the average result, where each time we randomly select a one-shot sample from the target domain.

4.3. Comparison with state-of-the-art methods

To evaluate our method, we compare existing methods in two scenarios: UDA and OSDA. Besides, we also provide extensive ablation studies to verify the effectiveness of the proposed methods.

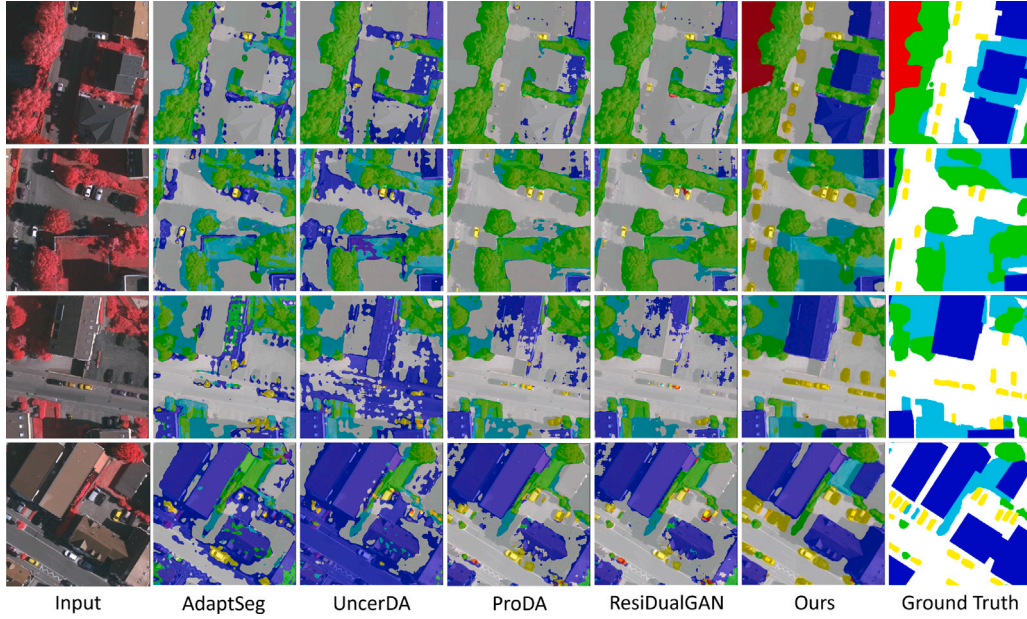
Unsupervised Domain Adaptation. We compare our method with the conventional UDA approaches where all target data is available. Specifically, we provide comparison results on Potsdam IR-R-G $\leftrightarrow$ Vaihingen IR-R-G in Tables 1 and 2, respectively. In this scenario, we compare recently cross-domain remote image segmentation approaches, including FSDAN [48], CIA-UDA [50], ResiDualGAN [49], and so on. To achieve a fair comparison, we provide performance based on different backbones, including ResNet-101 based on Deeplab-V3 and Mit-B5 based on SegFormer [57]. From the results, we can observe that our method provides a significant improvement compared to existing models. Specifically, our method achieves 71.61% mIoU and 64.49% mIoU results on Potsdam $\rightarrow$ Vaihingen and Vaihingen $\rightarrow$ Potsdam based on transformer architecture, which outperforms the baseline over 25.50% mIoU and 18.20% mIoU, respectively. Moreover, considering Deeplab-V3 architecture based on ResNet-101 backbone, our method achieves 64.89% mIoU result on Potsdam $\rightarrow$ Vaihingen, better than existing method ResiDualGAN [49] with 9.06% mIoU. The results demonstrate the effectiveness of our proposed MOAT on conventional unsupervised cross-domain semantic segmentation scenarios.

Moreover, we also provide the comparison results on LoveDA [44]. The detailed comparison results are shown in Table 3. We can observe

Table 3

Conventional cross-domain adaptation for RS image segmentation comparison results of LoveDA dataset. # TS denotes the number of target samples used in training. We use ResNet-50 as our backbone and the best results of domain adaptive segmentation are presented in **bold**.

Method	#TS	BG	Build.	Road	Water	Barren	Forest	Agricu.	mIoU	BG	Build.	Road	Water	Barren	Forest	Agricu.	mIoU
		Rural → Urban								Urban → Rural							
Source only	–	43.30	25.63	12.70	76.22	12.52	23.34	25.14	31.27	24.16	37.02	32.56	49.42	14.00	29.34	35.65	31.74
DDC [60]	All	43.60	15.37	11.98	79.07	14.13	33.08	23.47	31.53	25.61	44.27	31.28	44.78	13.74	33.83	25.98	31.36
AdaptSeg [1]	All	42.35	23.73	15.61	81.95	13.62	28.70	22.05	32.68	26.89	40.53	30.65	50.09	16.97	32.51	28.25	32.27
FADA [61]	All	43.89	12.62	12.76	80.37	12.70	32.76	24.79	31.41	24.39	32.97	25.61	47.59	15.34	34.35	20.29	28.65
CLAN [52]	All	43.41	25.42	13.75	79.25	13.71	30.44	25.80	33.11	22.93	44.78	25.99	46.81	10.54	37.21	24.45	30.39
TransNorm [62]	All	38.37	5.04	3.75	80.83	14.19	33.99	17.91	27.73	19.39	36.30	22.04	36.68	14.00	40.62	3.30	24.62
PyCDA [63]	All	38.04	35.86	45.51	74.87	7.71	40.39	11.39	36.25	12.36	38.11	20.45	57.16	18.32	36.71	41.90	32.14
CBST [5]	All	48.37	46.10	35.79	80.05	19.18	29.69	30.05	41.32	25.06	44.02	23.79	50.48	8.33	39.16	49.65	34.36
IAST [64]	All	48.57	31.51	28.73	86.01	20.29	31.77	36.50	40.48	29.97	49.48	28.29	64.49	2.13	33.36	61.37	38.44
PCEL [59]	All	54.19	51.54	47.83	77.99	37.99	23.80	38.35	47.38	–	–	–	–	–	–	–	–
MOAT (Ours)	All	96.52	35.15	48.45	39.08	66.27	30.62	40.81	50.60	90.63	48.57	32.88	35.35	49.06	6.47	29.51	40.54

**Fig. 3.** The visualization for predicted mask on the Potsdam → Vaihingen task.**Table 4**

One-shot cross-domain adaptation for Potsdam → Vaihingen and Vaihingen → Potsdam. # TS denotes the number of target samples used in training. We use ResNet-101 as our backbone and the best results are presented in **bold**.

Method	#TS	BA	BU	LV	TR	CA	IS	mIoU	BA	BU	LV	TR	CA	IS	mIoU
		Potsdam → Vaihingen							Vaihingen → Potsdam						
CycleGAN [65]	One	0.75	50.71	9.17	26.86	1.97	10.52	16.66	2.95	45.35	33.83	31.81	0.01	47.92	26.98
Benjdira et al. [66]	One	8.88	49.89	27.98	35.01	22.85	44.15	31.46	5.06	42.57	28.57	33.19	16.76	40.15	27.72
DualGAN [49]	One	1.91	35.51	6.73	43.15	7.06	6.50	16.81	2.27	34.15	44.35	21.48	28.95	37.58	28.13
ColorMapGAN [67]	One	10.12	49.33	25.37	36.60	2.41	24.03	24.64	2.42	17.39	35.46	1.22	0.44	37.97	15.82
LRM_I2I [15]	One	6.88	53.00	16.48	58.90	23.72	43.13	33.69	2.79	41.51	42.55	29.28	39.27	42.91	33.05
IDM [12]	One	20.47	57.92	45.30	41.28	31.90	50.61	41.25	8.31	49.47	56.39	36.27	44.19	49.97	40.77
MOAT (Ours)	One	23.18	59.50	43.21	45.73	32.92	49.37	42.32	7.15	51.74	57.40	38.00	46.01	51.36	41.94

that the proposed method **MOAT** provides a large margin improvement to the source-only baseline on both Rural → Urban and Urban → Rural tasks. Specifically, we achieve 50.6% mIoU on Rural → Urban and 40.54% mIoU on Urban → Rural, compared to 31.27% and 31.74%, respectively. Moreover, compared to representative domain adaption methods, our MOAT brings a significant improvement on conventional UDA scenarios, especially for 10.12% improvement on Rural → Urban compared to IAST [64].

Besides, we also provide the visualization results. In Fig. 3, we visualize the prediction results of Potsdam → Vaihingen task. We reproduce and compare existing methods, including AdaptSeg [1], UncerDA [3], ProDA [6], and ResiDualGAN [49]. The comparison results demonstrate our method's quality for image segmentation.

One-shot Unsupervised Domain Adaptation. To demonstrate the effectiveness of the proposed MOAT on one-shot domain adaptation, we conduct two different RS cross-domain segmentation tasks, including Potsdam ↔ Vaihingen and Rural ↔ Urban. The corresponding results are shown in Tables 4¹ and 5. We compare existing conventional UDA methods by modifying them to the one-shot domain adaptation setting, OSDA methods on natural image segmentation IDM [12], and OSDA method on RS image segmentation LRM_I2I [15].

¹ We denote the classes as clutter/background(BA), building (BU), low vegetation (LV), tree (TR), car (CA), and impervious surfaces (IS).

Table 5

One-shot cross-domain adaptation for RS image segmentation comparison results of LoveDA dataset. # TS denotes the number of target samples used in training. We use ResNet-50 as our backbone and the best results of one-shot domain adaptive segmentation are presented in **bold**.

Method	#TS	BG	Build.	Road	Water	Barren	Forest	Agricu.	mIoU	BG	Build.	Road	Water	Barren	Forest	Agricu.	mIoU
Rural → Urban										Urban → Rural							
AdaptSeg [1]	One	34.72	27.93	8.92	61.15	5.83	11.66	44.96	27.88	45.52	27.95	18.97	45.04	11.05	7.08	0.15	22.25
FADA [61]	One	35.52	5.44	1.37	49.65	0.19	9.17	39.80	19.73	35.91	6.86	10.75	15.23	2.91	6.22	0.11	11.13
CLAN [52]	One	20.51	27.36	10.77	55.75	7.84	28.10	35.03	26.48	44.44	41.35	14.31	27.43	8.27	3.49	0.81	20.01
TransNorm [62]	One	35.22	27.85	6.21	62.87	10.72	35.58	44.06	31.79	45.01	16.67	8.36	19.97	7.95	0.46	0	14.06
PyCDA [63]	One	27.09	37.29	14.20	33.08	35.20	18.66	34.78	28.61	34.43	13.95	17.76	43.78	6.06	0.23	19.92	19.45
CBST [5]	One	23.22	38.32	13.36	39.35	23.67	26.24	37.40	28.79	39.07	10.36	16.53	31.67	11.76	1.21	6.99	16.8
IAST [64]	One	32.48	32.77	16.62	54.21	14.34	33.62	18.68	28.96	51.48	22.58	22.05	41.16	7.63	1.56	29.04	25.07
IDM [12]	One	26.31	27.26	41.37	44.33	42.81	24.30	65.71	38.87	79.21	33.29	25.70	44.32	20.97	11.01	9.27	31.97
MOAT (Ours)	One	24.48	29.89	49.61	46.66	45.58	23.62	69.19	41.29	83.62	31.35	27.95	45.24	23.38	9.90	10.90	33.19

Table 6

Adaptation from GTA5 to Cityscapes based on ResNet101 backbone. # TS denotes the number of target samples used in training. We use Resnet-101 as our backbone and the best results of one-shot domain adaptation are presented in **bold**.

Method	#TS	road	side.	build.	wall	fence	pole	light	sign	vege.	terr.	sky	person	rider	car	truck	bus	train	motor.	bike	mIoU
GTA5 → Cityscapes																					
AdaptSeg [1]	One	77.7	19.2	75.5	11.7	6.4	16.8	18.2	15.4	77.1	34.0	68.5	55.3	30.9	74.5	23.7	28.3	2.9	14.4	18.9	35.2
CLAN [52]	One	77.1	22.7	78.6	17.0	14.8	20.5	23.8	12.0	80.2	39.5	74.3	56.6	25.2	78.1	29.3	31.2	0.0	19.4	16.7	37.7
ADVENT [4]	One	76.1	15.1	76.6	14.4	10.8	17.5	19.8	12.0	79.2	39.5	71.3	55.7	25.2	76.7	28.3	30.5	0.0	23.6	14.4	36.1
CBST [5]	One	76.1	22.2	73.5	13.8	18.8	19.1	20.7	18.6	79.5	41.3	74.8	57.4	19.9	78.7	21.3	28.5	0.0	28.0	13.2	37.1
ProDA [6]	One	80.9	32.2	68.9	24.7	21.0	24.6	29.6	14.8	71.7	28.6	66.4	55.8	17.5	81.6	21.2	24.2	20.0	25.0	13.9	38.0
ASM [11]	One	89.5	31.2	81.3	27.8	22.8	30.6	32.8	25.1	82.6	35.0	76.7	59.2	26.6	82.3	27.7	34.1	0.9	25.6	29.6	43.2
SM-PPM [54]	One	85.0	23.2	80.4	21.3	24.5	30.0	32.0	26.7	83.2	34.8	74.0	57.3	29.0	77.7	27.3	36.5	5.0	28.2	39.4	42.8
IDM [12]	One	89.7	43.5	83.3	27.1	24.2	33.4	37.2	24.2	80.2	19.6	83.9	67.1	36.4	86.2	34.2	41.6	2.7	26.9	17.9	45.2
MOAT (Ours)	One	84.5	29.8	81.6	31.1	30.0	30.4	35.5	19.7	78.9	37.9	77.1	64.1	33.8	87.9	48.2	32.5	19.6	36.4	16.4	46.1
SYNTHIA → Cityscapes																					
AdaptSeg [1]	One	64.1	25.6	75.3	–	–	–	4.7	2.7	77.0	–	70.0	52.2	20.6	51.3	–	22.4	–	19.9	22.3	39.1
CLAN [52]	One	68.3	26.9	72.2	–	–	–	5.1	5.3	75.9	–	71.4	54.8	18.4	65.3	–	19.2	–	22.1	20.7	40.4
ADVENT [4]	One	65.7	22.3	69.2	–	–	–	2.9	3.3	76.9	–	69.2	55.4	21.4	77.3	–	17.4	–	21.4	16.7	39.9
CBST [5]	One	59.6	24.1	72.9	–	–	–	5.5	13.8	72.2	–	69.8	55.3	21.1	57.1	–	17.4	–	13.8	18.5	38.5
ProDA [6]	One	81.8	38.9	60.6	–	–	–	14.6	11.5	51.5	–	69.9	56.2	16.4	79.2	–	24.4	–	5.9	32.3	41.8
ASM [11]	One	85.7	39.7	77.1	–	–	–	2.1	9.2	76.9	–	81.7	43.4	11.4	63.9	–	15.8	–	1.6	20.3	40.7
SM-PPM [54]	One	79.3	35.3	75.9	–	–	–	25.4	22.7	79.9	–	76.8	54.6	23.5	60.2	–	23.9	–	21.2	36.6	47.3
IDM [12]	One	84.9	36.8	78.5	–	–	–	4.5	5.3	73.1	–	88.3	60.4	27.8	69.6	–	33.9	–	23.4	35.4	47.8
MOAT (Ours)	One	79.4	29.1	82.1	–	–	–	11.5	13.2	79.4	–	86.5	60.4	28.4	78.5	–	32.4	–	25.0	36.9	49.5

For comparison, we can observe that the proposed MOAT performs significantly better than existing methods. For detail, our MOAT achieves 43.2% and 41.94% mIoU for Potsdam → Vaihingen and Vaihingen → Potsdam on the one-shot UDA setting, which provides a significant improvement compared to existing methods. As for Rural/Urban → Urban/Rural, our method brings almost 2.42%/1.22% mIoU with achieving 41.29%/33.19% results.

Besides RS cross-domain image segmentation, we also demonstrate the effectiveness of our method on the synthetic-to-real scenario. As Table 6 shows, we compare existing one-shot UDA methods on GTA5 → Cityscapes and SYNTHIA → Cityscapes. We can observe that our method achieves 46.1%/49.5% mIoU results on GTA5→Cityscapes and SYNTHIA→Cityscapes, respectively, providing a significant margin improvement over ASM (43.2%/40.7%) and SM-PPM (42.8%/47.3%). It demonstrates the effectiveness of the proposed minimax one-shot adaptation approach. On the other hand, the performance of the conventional UDA method degrades significantly in such a data-scarce scenario.

4.4. Ablation study

In this section, we perform ablation studies of the proposed method on the remote sensing dataset LoveDA. The influence of different components is elaborated below.

Influence of Different Components. We conduct experiments to verify the effectiveness of the proposed components on RS cross-domain image segmentation task from Rural → Urban and Urban → Rural, including the uncertainty-based target distribution estimation (TDE), sample selection in maximization (SSM), and memory-enhanced

Table 7

Study each component adopted by our MOAT on Rural → Urban and Urban → Rural. TDE: target distribution estimation, MemPro: memory enhanced prototype learning, and SSM: sample selection in maximization.

Method	TDE	MemPro	SSM	R → U	U → R
Baseline				31.3± 0.1	31.7± 0.3
(a)	✓			36.7 ± 0.9	32.2 ± 0.3
(b)	✓	✓		37.5 ± 0.6	32.9 ± 0.2
(c)	✓		✓	38.2 ± 0.7	33.0 ± 0.6
(d)		✓		36.1 ± 0.7	32.5 ± 0.8
(e)		✓	✓	36.9± 0.4	32.7 ± 0.6
MOAT	✓	✓	✓	41.3±0.5	33.2 ± 0.3

prototype learning (MemPro). We use the source-only method as our baseline model. As Table 7 shows, one the task Rural → Urban, our model provides a significant improvement by achieving 41.3% mIoU, compared to the baseline model of 31.3%. Specifically, model with TDE (model (a)) improves performance from 31.3% to 36.7%, which indicates the proposed target distribution estimation with uncertainty can effectively transfer the target information into the trained model. Furthermore, it should be noticed that adding SSM to MemPro produces a marginal improvement, compared with model (d) and model (e), by achieving 36.9% from 36.1%. However, it becomes significant based on TDE, compared with model (c) and model (a) with 38.2% v.s. 36.7%.

Performance of Different Backbones. In this subsection, we provide the performance of our method based on different backbones, including Resnet101 from DeepLab-V3 and Transformer from DAFormer. We perform experiments on Potsdam ↔ Vaihingen tasks. The detailed

Table 8

Comparison results on different backbones. Our method is based on DeepLab V2 (DLv2) using ResNet-101 as the backbone and DAFormer using Transformer as the backbone.

	Potsdam → Vaihingen		Vaihingen → Potsdam	
	One-shot UDA	UDA	One-shot UDA	UDA
Source Only _{DLv2}	34.81	34.81	32.54	32.54
MOAT_{DLv2}	42.32	64.89	41.94	54.58
Source Only _{SegFormer}	46.11	46.11	46.29	46.29
MOAT_{SegFormer}	53.72	71.61	51.35	64.49

Table 9

Study on the beta distribution parameter λ_B .

λ_B	mIoU
0.1	36.2 $\pm$ 0.6
0.5	36.7 $\pm$ 0.6
0.6	36.3 $\pm$ 0.4
0.7	36.5 $\pm$ 0.5
1.0	36.0 $\pm$ 0.4

Table 10

Study on the number of selected samples K .

K	mIoU
1	36.2 $\pm$ 0.3
2	37.8 $\pm$ 0.2
3	38.2 $\pm$ 0.3
4	38.2 $\pm$ 0.2
5	37.9 $\pm$ 0.3

Table 11

Study on memory momentum parameter m .

m	mIoU
0.999	40.9 $\pm$ 0.5
0.99	41.0 $\pm$ 0.2
0.9	40.9 $\pm$ 0.4
0.8	41.3 $\pm$ 0.4
0.7	40.7 $\pm$ 0.3

results are shown in Table 8. The results demonstrate that our method provides a significant improvement in different backbones compared to source-only model, for both unsupervised domain adaptation (UDA) and one-shot domain adaptation (One-shot UDA) scenarios.

4.5. Parameters analysis

Our framework contains several new hyper-parameters, including the beta distribution's shape parameter λ_B in Eq. (3), the number of selected stylized training images K as well as n_s in Eq. (5), and the momentum m on prototype memory update in Eq. (6). We construct extensive experiments on Rural → Urban to analyze these hyper-parameters and observe that the results of our MOAT are not sensitive to them. The detailed results and analysis are as follows.

We first analyze the hyper-parameter λ_B , the beta distribution shape parameter that controls the distribution of the convex in Eq. (3). We choose the beta distribution rather than the Gaussian distribution to bias the value from 0 to 1. The results are shown in Table 9. We can observe that the accuracy arrives at a peak when $\lambda_B = 0.5$. The results suggest that the performance is not too sensitive to λ_B , and choosing $\lambda_B = 0.5$ seems to be a good starting point.

Another important hyper-parameter is the number of selected stylized samples K in Eq. (5), which means K selected samples can describe the target distribution. The results in Table 10 suggest that the best performance is when $K = 3$ or $K = 4$. Due to K also influencing the convergence speed, we choose $K = 3$ in all experiments.

Table 12

Study on the hyper-parameters n_s on the GTA5 → Cityscapes adaptation.

n_s	mIoU	Time (s)
100	46.1	675
200	46.7	988
300	46.8	1223
400	46.9	1460
500	46.9	1824
1000	47.1	2370

As for the hyper-parameter momentum m in Eq. (6), it affects the proportion of stylized features in the prototype memory. A large m means the source feature dominates the prototype memory, and a small m represents that the influence of the stylized feature is significant. The results are shown in Table 11. We can observe that our method is stable to m and the best result is performed when $m = 0.8$.

The last hyper-parameter n_s denotes the size of the source stylized images pool in the sample selection maximization process, which influences the quality of estimating the target distribution from selected images. Generally, the larger n_s is, the higher accuracy of the model obtained. To demonstrate this, we have also conducted the corresponding experiments to investigate the impact of varying different values of n_s on the model's accuracy. Specifically, we conducted experiments on GTA5→Cityscapes, and we observed the accuracy and the training time (s) of one epoch. The results are shown in Table 12. From the results, we can notice that a larger number of n_s often leads to higher accuracy in estimating the target distribution. However, the training time also increases as n_s becomes larger. To quickly adapt to dynamic environments, we select a smaller n_s due to the speed and accuracy of the trade-off.

4.6. Convergence speed analysis

To verify the quick adaption of our method, we have conducted experiments to compare the convergence speed during adaptation training. We conduct experiments on GTA5→Cityscapes and compare them with SM-PPM [54]. As Fig. 4 shows, our method can quickly fit into the new target domain with only about 50 training iterations. It reveals that our MOAT is superior in both efficiency and effectiveness.

5. Conclusions

This paper introduces a one-shot domain adaptation framework, named **MOAT**, transferring a pre-trained source model to a target domain for remote sensing semantic segmentation. During adaptation, we only use a single target image. The core idea involves alternately maximizing cross-entropy to identify the most informative source images that can identify the target distribution, and minimizing the cross-entropy of the selected samples to enable the model to fit the target data. Additionally, the proposed memory-based prototype enhancement strategy learns domain-invariant decision boundaries. The provided one-shot domain adaptation method is important and practical for data-restricted scenarios, especially for data-privacy remote sensing image processing. The effectiveness of the proposed MOAT has been demonstrated by achieving a new state-of-the-art performance on two standard one-shot domain adaptive remote sensing image semantic segmentation benchmarks.

Limitations and Future Works. Although the proposed MOAT framework is effective for one-shot domain adaptation, there are still some limitations that need to be addressed. For instance, its reliance on a single target data point might limit its generalizability to more complex target distributions, and it faces challenges in handling multiple domains and dynamic domain shifts in real-world scenarios. In future work, we will address these challenges from the following three aspects. First, we will enhance the proposed method by leveraging

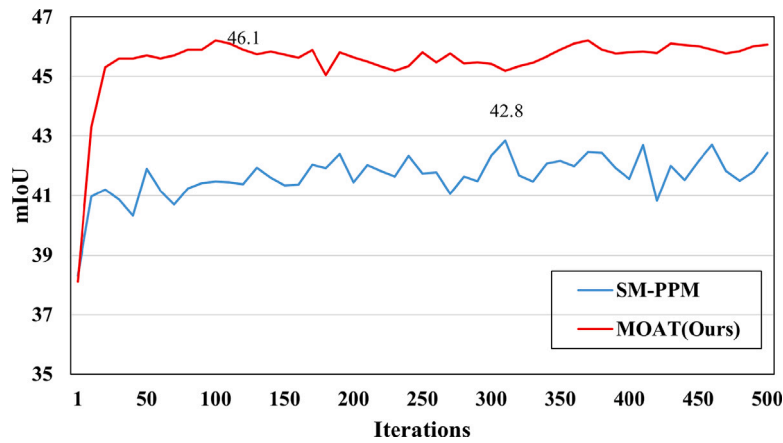


Fig. 4. The visualization of convergence and training iterations.

powerful pre-trained generative models (e.g., Stable Diffusion Model) to obtain diverse pseudo-target images. Second, we will extend the MOAT framework to handle adaptation across multiple domains, addressing more complex and dynamic domain shifts. Lastly, we will investigate advanced optimization techniques to enhance the efficiency and effectiveness of the adaptation process, potentially reducing computational requirements and improving adaptability.

CRedit authorship contribution statement

Jiaojiao Tan: Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Haiwei Zhang:** Writing – review & editing, Methodology. **Ning Yao:** Resources, Methodology. **Qiang Yu:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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